

Machine Learning Answers (100% Correct)

Q1:

Explain the concept of backpropagation in neural networks. (10 Marks)

Answer:

Backpropagation is the core learning algorithm used to train neural networks. During forward propagation, input data passes through network layers to produce an output. The predicted output is compared with the actual target using a loss function to calculate error. Backpropagation then computes gradients of the loss with respect to each weight using the chain rule. These gradients are used with optimization methods such as gradient descent to update the weights, allowing the network to reduce error and improve predictions over time.

Q2:

What is overfitting in machine learning? How can it be prevented? (5 Marks)

Answer:

Overfitting occurs when a machine learning model memorizes the training data rather than learning general patterns. As a result, the model performs well on training data but poorly on new unseen data. Overfitting can be prevented using techniques such as regularization (L1 or L2), dropout in neural networks, early stopping, cross-validation, and increasing the amount or diversity of training data.

Q3:

Define gradient descent and explain its role in training neural networks. (5 Marks)

Answer:

Gradient descent is an optimization algorithm used to minimize the loss function of a model. It works by computing the gradient (derivative) of the loss function with respect to model parameters and updating the parameters in the direction opposite to the gradient. This iterative process gradually reduces error during training. Variants include batch gradient descent, stochastic gradient descent (SGD), and mini-batch gradient descent.

Q4:

What is the difference between classification and regression in machine learning? (5 Marks)

Answer:

Classification and regression are two types of supervised learning tasks. Classification predicts discrete class labels, such as spam vs non-spam emails or disease vs no disease. Regression predicts continuous numerical values, such as house prices or temperature. Typical classification algorithms include logistic regression, decision trees, and support vector machines, while regression tasks often use linear regression or polynomial regression.

